

Polynomials – 1

Roots, multiplicity and the derivative

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Theoretical minimum.

Theorem 1 (multiplicity and the derivative). A number c is a root of f of multiplicity k if and only if $f(c) = f'(c) = \dots = f^{(k-1)}(c) = 0 \neq f^{(k)}(c)$. Consequently $f/\gcd(f, f')$ has the same roots as f , all simple, and f is a constant times a k -th power exactly when every root has multiplicity divisible by k .

Theorem 2 (Rolle for polynomials). If a real polynomial of degree n has n real roots counted with multiplicity, then so does f' , and its roots interlace those of f . In particular the roots of f' lie in the convex hull of the roots of f , which over $\mathbb{C}$ is the Gauss–Lucas theorem.

Fact 1 (Vieta and positivity). The coefficients are the elementary symmetric functions of the roots up to sign. A real polynomial is non-negative on $\mathbb{R}$ if and only if it is a sum of two squares of real polynomials; the discriminant of a quadratic decides the sign pattern, and for higher degrees the sign of f between consecutive roots alternates.

1. (Putnam 1940) Let $f(x)$ be a polynomial of degree n such that a power of $f(x)$ is divisible by a power of its derivative, that is, $(f(x))^p$ is divisible by $(f'(x))^q$ for some positive integers p, q . Prove that $f(x)$ is divisible by $f'(x)$ and that $f(x)$ has a single root of multiplicity n .
2. (Putnam 1941) Let the roots $a \leq b \leq c$ of $f(x) = x^3 + px^2 + qx + r$ be real. Prove that if the interval (b, c) is divided into six equal parts, then a root of $f'(x) = 0$ lies in the fourth part counting from b . What is the form of f when that root falls at either end of the fourth part?

3. (Putnam 1999) Let $P(x)$ be a polynomial of degree n such that $P(x) = Q(x)P''(x)$, where $Q(x)$ is a quadratic polynomial. Show that if P has at least two distinct roots, then it has n distinct roots. The roots may be real or complex.
4. (Putnam 1991) Find all real polynomials $p(x)$ of degree $n \geq 2$ for which there are real numbers $r_1 < r_2 < \cdots < r_n$ with $p(r_i) = 0$ for all i and $p'(\frac{r_i + r_{i+1}}{2}) = 0$ for $i = 1, \dots, n-1$.
5. (Putnam 1999) Let $p(x)$ be a polynomial that is non-negative for all real x . Prove that for some k there are polynomials $f_1, \dots, f_k$ with $p(x) = \sum_{j=1}^k (f_j(x))^2$.